Discrete Mathematics

Final Exam Preparation Pack

Quick Revision + 50 Practice Questions

PART 1	PART 2	PART 3
Quick Revision	**MCQ (25 Questions)**	**Written (25 Questions)**
Page 1 — Proofs	Q1–Q8 Set Theory	Q26–Q34 Set Theory
Page 2 — Set Theory	Q9–Q16 Proofs	Q35–Q42 Proofs
Page 3 — Logic & Circuits	Q17–Q25 Logic	Q43–Q50 Logic

Topics Covered: Sets · Operations · Cardinality · Countability · Russell's Paradox · Cantor's Diagonal · Direct Proof · Contrapositive · Contradiction · Cases · Existence · Uniqueness · Vacuous · Propositional Logic · Operators · Equivalence · Predicates · Quantifiers · Inference · CNF/DNF · Circuits · K-Map

Medium | **Medium-Hard** | **Hard**

Difficulty: starts medium, builds to hard

Good luck on your exam! ★

Read each question carefully · Show all working in written questions · Manage your time

§1 QUICK REVISION — Proof Methods

GOLDEN RULE: State your method · Assume the hypothesis · Justify every step · End with ■

Direct Proof

Goal: Prove $P \Rightarrow Q$
Method: Assume P, derive Q step by step.

Key patterns:

- Even n: write $n = 2k$
- Odd n: write $n = 2k + 1$
- $a \mid b$: write $b = ak$, $k \in \mathbb{Z}$
- Rational r: write $r = p/q$, $\gcd(p, q) = 1$

Example: n even $\Rightarrow n^2$ even
$n = 2k \Rightarrow n^2 = 4k^2 = 2(2k^2)$ ✓

Contrapositive Proof

Goal: Prove $P \Rightarrow Q$
Method: Prove $\neg Q \Rightarrow \neg P$ instead

$P \Rightarrow Q \equiv \neg Q \Rightarrow \neg P$

Use when: $\neg Q$ is a stronger, more useful hypothesis.

Example: n^2 even $\Rightarrow n$ even
Contra: n odd $\Rightarrow n^2$ odd
$n = 2k+1 \Rightarrow n^2 = 4k^2+4k+1 = 2(\ldots)+1$ ✓

Proof by Contradiction

Goal: Prove statement S
Method: Assume $\neg S$, derive $C \wedge \neg C$

Classic examples:

- $\sqrt{2}$ irrational: assume $\sqrt{2} = p/q$, derive both p, q even (contradicts $\gcd = 1$)
- Infinitely many primes: assume finite list, construct $N = p_1 \cdots p_n + 1$, contradiction

Proof by Cases

Method: Split domain into exhaustive cases; prove Q in each.

Common splits:
Even / Odd | $n \bmod 3 \in \{0, 1, 2\}$
Positive / Zero / Negative

Example: $n^2 + n$ always even
Case 1: $n = 2k \Rightarrow n^2 + n = 2k(2k + 1)$ even
Case 2: $n = 2k+1 \Rightarrow n^2+n = 2(\ldots)$ even

Existence & Uniqueness

Constructive existence: exhibit specific x_0; verify $P(x_0)$.
Non-constructive: argue existence via contradiction or cases.

Uniqueness: assume x_1, x_2 both satisfy P; prove $x_1 = x_2$.

Must have BOTH parts for $\exists$!

Vacuous Proof

$P \Rightarrow Q$ is **vacuously true** when P is always false.

Because: $P \Rightarrow Q \equiv \neg P \vee Q$
If P is always false, $\neg P$ is true, whole thing true.

Example: "If $x^2 < 0$ then $x = 7$" — vacuously true for $x \in \mathbb{R}$.

Key: Look for impossible hypothesis.

Method	Assume	Derive	Conclude
Direct	P	$\cdots \Rightarrow Q$	Q **holds**
Contrapositive	$\neg Q$	$\cdots \Rightarrow \neg P$	$P \Rightarrow Q$ **holds**
Contradiction	$\neg S$	**contradiction**	S **holds**
Cases	$C_1; C_2; \ldots$	Q **in each**	Q **always holds**
Vacuous	—	P **is impossible**	$P \Rightarrow Q$ **trivially true**

§2 QUICK REVISION — Set Theory, Cardinality & Countability

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Sets — Core Definitions

Set: Unordered collection of distinct elements.

$x \in A$ (element) $A \subseteq B$ (subset)
$A \subset B$ (proper subset) $A = B$ iff $A \subseteq B$ and $B \subseteq A$

Power Set: $\mathcal{P}(A)$ = all subsets of A
$|A| = n \Rightarrow |\mathcal{P}(A)| = 2^n$

Special sets: $\emptyset \subseteq$ every set
$\mathbb{N} \subset \mathbb{W} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R}$

Set Operations

Union	$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$
Intersection	$A \cap B = \{x \mid x \in A \text{ and } x \in B$
Difference	$A \setminus B = \{x \in A \mid x \notin B\}$
Complement	$\bar{A} = U \setminus A$
Sym. Diff.	$A \triangle B = (A \setminus B) \cup (B \setminus A)$
Cartesian	$A \times B = \{(a, b) \mid a \in A, b \in B$

$|A \cup B| = |A| + |B| - |A \cap B|$ (inclusion-exclusion)

Key Laws

De Morgan:
$\overline{A \cup B} = \bar{A} \cap \bar{B}$ $\overline{A \cap B} = \bar{A} \cup \bar{B}$

Distributive:
$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Absorption:
$A \cup (A \cap B) = A$ $A \cap (A \cup B) = A$

Double complement: $\bar{\bar{A}} = A$

Russell's Paradox

Let $R = \{x \mid x \notin x\}$
If $R \in R$: satisfies condition $\Rightarrow R \notin R$ **(!!)**
If $R \notin R$: satisfies condition $\Rightarrow R \in R$ **(!!)**

Resolution: ZFC axiomatic set theory forbids unrestricted self-reference. Sets can only be formed from existing sets: $\{x \in A \mid P(x)\}$.

Cardinality & Countability

$|A| = |B|$ iff $\exists$ bijection $f : A \to B$

Set	Cardinality	Type
$\mathbb{N}, \mathbb{W}, \mathbb{Z}, \mathbb{Q}$	$\aleph_0$	Countably ∞
$\mathbb{R}$	$\mathfrak{c} = 2^{\aleph_0}$	Uncountable

Bijections to $\mathbb{N}$:
$\mathbb{Z}$: interleave $0, 1, -1, 2, -2, \ldots$
$\mathbb{Q}$: Cantor's diagonal zig-zag grid

Cantor's Diagonalization

Claim: $[0, 1]$ is uncountable.

Proof sketch: Assume a list $r_1, r_2, \ldots$ covers all reals.
Build s with $s_i \neq d_{ii}$ (diagonal digit).
Then $s \neq r_i$ for every i — not in list! **(!!)**

Same trick: Halting Problem undecidability.

Cantor's General Theorem:
$|A| < |\mathcal{P}(A)|$ for every set A
$\aleph_0 < 2^{\aleph_0} < 2^{2^{\aleph_0}} < \cdots$

To prove $A = B$:	Show $A \subseteq B$ AND $B \subseteq A$	(element-chasing)
To prove $A \subseteq B$:	Take arbitrary $x \in A$, show $x \in B$	(universal approach)
To prove $\lvert A\rvert = \lvert B\rvert$:	Exhibit explicit bijection $f : A \to B$	(for infinite sets)
To prove A uncountable:	Diagonalize — construct element outside any list	

§3 QUICK REVISION — Logic, Quantifiers, Circuits & K-Maps

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Logical Operators & Truth Table

p	q	$\neg p$	$p \wedge q$	$p \vee q$	$p \to q$	$p \leftrightarrow q$
T	T	F	T	T	T	T
T	F	F	F	T	**F**	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Precedence: $\neg > \wedge > \vee > \to > \leftrightarrow$

$p \to q \equiv \neg p \vee q$ (crucial!)

Key Equivalences

De Morgan: $\neg(p \wedge q) \equiv \neg p \vee \neg q$
Implication: $p \to q \equiv \neg p \vee q$
Contrapositive: $p \to q \equiv \neg q \to \neg p$
Double Neg: $\neg\neg p \equiv p$
Absorp: $p \vee (p \wedge q) \equiv p$
Distrib: $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$

Tautology: $p \vee \neg p$ **Contradiction:** $p \wedge \neg p$

Predicates & Quantifiers

$P(x)$ becomes proposition when x assigned.

$\forall x\, P(x)$: false if ONE counterexample
$\exists x\, P(x)$: true if ONE witness

Negation (De Morgan for quant.):
$\neg\forall x\, P(x) \equiv \exists x\, \neg P(x)$
$\neg\exists x\, P(x) \equiv \forall x\, \neg P(x)$

Order matters: $\forall x\, \exists y \neq \exists y\, \forall x$

Rules of Inference

Modus Ponens	$p,\ p \to q\ \therefore q$
Modus Tollens	$\neg q,\ p \to q\ \therefore \neg p$
Hyp. Syllogism	$p \to q,\ q \to r\ \therefore p \to r$
Disj. Syllogism	$p \vee q,\ \neg p\ \therefore q$
Addition	$p\ \therefore p \vee q$
Simplification	$p \wedge q\ \therefore p$
Conjunction	$p, q\ \therefore p \wedge q$
Resolution	$p \vee q,\ \neg p \vee r\ \therefore q \vee r$

Normal Forms (CNF / DNF)

CNF: AND of ORs — $(l_1 \vee l_2) \wedge (l_3 \vee l_4)$
DNF: OR of ANDs — $(l_1 \wedge l_2) \vee (l_3 \wedge l_4)$

To CNF: eliminate $\to$; push $\neg$ inward (De Morgan); distribute $\wedge$ over $\vee$.
To DNF: distribute $\vee$ over $\wedge$ OR read from truth table (pick T-rows).

Literal: p or $\neg p$

Circuits & K-Map

Gates: AND($\cdot$), OR($+$), NOT($^{-}$), NAND, NOR, XOR($\oplus$)
Universal: NAND or NOR alone can build any circuit.

K-Map rules:

- Groups must be powers of 2 (1,2,4,8,16)
- Largest possible groups; overlapping OK
- Gray code column order: 00,01,11,10
- Edges wrap around (toroid topology)
- Each group of 2^k cells eliminates k literals

SAT: at least one T in truth table **Tautology:** all T in truth table **Contradiction:** all F in truth table

Nested Quantifier Negation: push $\neg$ inward, flip each quantifier, negate innermost predicate

$\neg \forall x \, \exists y \, P(x, y) \;\equiv\; \exists x \, \forall y \, \neg P(x, y)$

PART 2 — 25 Multiple Choice Questions Q1–Q25

Instructions: Circle the best answer. Each question has exactly one correct answer. **Marks:** 2 each **Total: 50 marks**

Set Theory & Cardinality Q1–Q8

Q1. Let $A = \{1, 2, 3, 4\}$ and $B = \{3, 4, 5, 6\}$. What is $|A \triangle B|$?
(a) 2 (b) 4 **(c) 4** (d) 6
Hint: $A \triangle B = \{1, 2, 5, 6\}$ — elements in exactly one of A, B.

Q2. If $|A| = 5$ and $|B| = 4$ and $|A \cap B| = 2$, what is $|A \cup B|$?
(a) 7 **(b) 7** (c) 9 (d) 11
$|A \cup B| = 5 + 4 - 2 = 7$

Q3. Which of the following is always true for any sets A and B?
(a) $A \cap B = A \cup B$ **(b)** $A \cap B \subseteq A \subseteq A \cup B$
(c) $A \subseteq A \cap B$ (d) $A \cup B \subseteq A$

Q4. The power set of $\{a, b\}$ is:
(a) $\{\{a\}, \{b\}\}$ **(b)** $\{\emptyset, \{a\}, \{b\}, \{a, b\}\}$
(c) $\{a, b, \emptyset\}$ (d) $\{\{a, b\}, \emptyset\}$

Q5. Which statement about $\emptyset$ is **FALSE**?
(a) $\emptyset \subseteq \{1, 2, 3\}$ (b) $\emptyset \subseteq \emptyset$ **(c)** $\emptyset \in \emptyset$ (d) $|\mathcal{P}(\emptyset)| = 1$

Q6. Which set has the same cardinality as $\mathbb{N}$?
(a) $\mathbb{R}$ (b) $[0, 1]$ **(c)** $\mathbb{Q}$ (d) $\mathcal{P}(\mathbb{N})$

Q7. In Russell's Paradox, the set $R = \{x \mid x \notin x\}$ leads to a contradiction because:
(a) R is too large to be a set
(b) $R \in R$ and $R \notin R$ are both forced simultaneously
(c) R is the empty set
(d) R contains only one element

Q8. Cantor's diagonalization argument proves that the real numbers in $[0, 1]$ are:
(a) Countably infinite **(b) Uncountably infinite**
(c) Finite (d) In bijection with $\mathbb{Q}$
Because any supposed list $r_1, r_2, \ldots$ misses the constructed diagonal number s.

Proof Methods Q9–Q16

Q9. To prove "if n^2 is odd then n is odd", the most efficient method is:
(a) Direct proof **(b) Contrapositive** (c) Contradiction (d) Vacuous

Q10. Which of the following correctly begins a proof by contradiction of the statement S?
(a) "Assume S is true..."
(b) "Suppose for contradiction that $\neg S$..."
(c) "We will prove the contrapositive..."
(d) "Let us consider two cases..."

Q11. A proof that "$2x + 3 = 9$ has a unique solution" must include:
(a) Only showing $x = 3$ works
(b) Only showing no other value works
(c) Both showing $x = 3$ works AND showing any two solutions must be equal
(d) A contradiction argument

Q12. The statement "For all real x, if $x^2 + 4 = 0$ then $x = 100$" is:
(a) False, since $x = 100$ doesn't satisfy $x^2 + 4 = 0$
(b) Vacuously true, since $x^2 + 4 \geq 4 > 0$ for all real x
(c) True by contradiction
(d) Unprovable

Q13. Which of the following is NOT a valid step in a direct proof that $m+n$ is even given m, n both even?
(a) Write $m = 2a$ and $n = 2b$
(b) Compute $m + n = 2a + 2b = 2(a + b)$
(c) Assume $m + n$ is odd and derive a contradiction
(d) Conclude $m + n$ is even since $a + b \in \mathbb{Z}$

Q14. When proving $P \Rightarrow Q$ by the contrapositive, which of the following is the correct setup?
(a) Assume P is false; derive Q is false
(b) Assume Q is false; derive P is false
(c) Assume both P and Q are false
(d) Assume P is true; derive Q is false, then find a contradiction

Q15. In the proof that $\sqrt{2}$ is irrational, after writing $\sqrt{2} = p/q$ with $\gcd(p, q) = 1$ and squaring to get $2q^2 = p^2$, the next conclusion is:
(a) p must be even (b) q must be even
(c) p and q are equal (d) $p/q < 1$

Q16. Proof by cases requires that the cases be:
(a) Exactly two in number
(b) Mutually exclusive only

(c) Exhaustive — together covering all possibilities
(d) Each proved using contradiction

Logic, Quantifiers, Inference, Circuits Q17–Q25

Q17. Which assignment makes $(p \to q) \wedge (q \to p)$ FALSE?
(a) $p = T, q = F$ (b) $p = F, q = F$ (c) $p = T, q = T$ (d) $p = F, q = T$
$p \to q$ is F when $p = T, q = F$; so the whole conjunction is F.

Q18. $\neg(p \wedge q) \wedge (p \vee q)$ simplifies to:
(a) $p \oplus q$ **(XOR)** (b) $p \wedge q$ (c) $\neg p \wedge \neg q$ (d) Tautology

Q19. The negation of $\exists x\, \forall y\, (P(x) \to Q(y))$ is:
(a) $\forall x\, \exists y\, (\neg P(x) \to \neg Q(y))$
(b) $\forall x\, \exists y\, (P(x) \wedge \neg Q(y))$
(c) $\exists x\, \forall y\, \neg P(x)$
(d) $\forall x\, \forall y\, \neg Q(y)$

Q20. Which rule of inference is used here? "If it is sunny, we go to the park. We do not go to the park. Therefore it is not sunny."
(a) Modus Ponens **(b) Modus Tollens**
(c) Disjunctive Syllogism (d) Hypothetical Syllogism

Q21. Which formula is in **Conjunctive Normal Form (CNF)**?
(a) $(p \wedge q) \vee r$ **(b)** $(p \vee q) \wedge (\neg p \vee r)$
(c) $p \vee (q \wedge r)$ (d) $p \wedge q \wedge \neg r \wedge s$

Q22. A 3-variable K-Map has how many cells?
(a) 4 **(b) 8** (c) 6 (d) 16

Q23. In K-Map minimisation, grouping 4 cells in a 4-variable map eliminates how many literals from the product term?
(a) 1 **(b) 2** (c) 3 (d) 4

Q24. Which gate is described as "universal" — meaning any logic function can be built from it alone?
(a) AND (b) XOR **(c) NAND** (d) OR

Q25. The argument "All mammals breathe air. Whales breathe air. Therefore whales are mammals." is:
(a) Valid — Modus Ponens
(b) Invalid — fallacy of affirming the consequent

(c) Valid — Hypothetical Syllogism
(d) Valid — Disjunctive Syllogism

PART 3 — 25 Written Questions Q26–Q50

Instructions: Show all working clearly. State any proof method used. **Marks:** variable (see each question) **Total: 75 marks**

Set Theory & Cardinality Q26–Q34

Q26. *[3 marks]* Let $A = \{x \in \mathbb{Z} \mid x^2 \leq 9\}$ and $B = \{-1, 0, 1, 2, 3, 4\}$.
(a) List all elements of A.
(b) Find $A \cap B$ and $A \setminus B$.
(c) Is $A \subseteq B$? Justify.

Q27. *[3 marks]* For sets A, B, C in universe U, prove using set identities:

$$A \setminus (B \cup C) = (A \setminus B) \cap (A \setminus C)$$

(Hint: use De Morgan and $A \setminus X = A \cap \bar{X}$.)

Q28. *[2 marks]* Let $A = \{1, 2, 3\}$. List all elements of $\mathcal{P}(A)$ and verify $|\mathcal{P}(A)| = 2^{|A|}$.

Q29. *[3 marks]* In a class of 60 students: 35 study Maths, 28 study CS, and 15 study both.
(a) How many study Maths or CS?
(b) How many study exactly one subject?
(c) How many study neither?

Q30. *[3 marks]* Prove or disprove: For any sets A and B,

$$A \times (B \cap C) = (A \times B) \cap (A \times C)$$

Q31. *[2 marks]* Explain Russell's Paradox in your own words. What does it reveal about naive set theory, and how does ZFC resolve it?

Q32. *[3 marks]* Show that $|\mathbb{Z}| = |\mathbb{N}|$ by constructing an explicit bijection $f : \mathbb{N} \to \mathbb{Z}$ and proving it is a bijection (one-to-one and onto).

Q33. *[4 marks]* Use Cantor's diagonalization argument to prove that the set of all infinite binary strings is uncountable.
(Hint: assume a list $s_1, s_2, s_3, \ldots$ of all binary strings; construct a new string differing from each s_i at position i.)

Q34. *[3 marks]* Determine (with justification) whether each of the following sets is countable or uncountable:

(a) The set of all polynomials with integer coefficients
(b) The set of all functions from $\{0, 1\}$ to $\mathbb{N}$
(c) The set of all functions from $\mathbb{N}$ to $\{0, 1\}$

Proof Methods Q35–Q42

Q35. *[3 marks]* Prove directly: If n is an integer such that $n \equiv 1 \pmod 3$, then $n^2 \equiv 1 \pmod 3$.
(Write $n = 3k + 1$ and expand.)

Q36. *[3 marks]* Prove by contrapositive: If n^3 is even, then n is even.

Q37. *[4 marks]* Prove by contradiction that $\sqrt{3}$ is irrational.
(Model your proof on the $\sqrt{2}$ argument; use the fact that $3 \mid p^2 \Rightarrow 3 \mid p$.)

Q38. *[3 marks]* Prove by cases that for any integer n: $n^2 + n + 1$ is always odd.
(Consider n even and n odd separately.)

Q39. *[3 marks]* Prove: For any integers a, b, c, if $a \mid b$ and $a \mid c$ then $a \mid (mb + nc)$ for any integers m, n.
(This is the "linear combination" divisibility result.)

Q40. *[3 marks]* Prove the existence and uniqueness of the solution to $3x - 6 = 9$.
(Show existence by finding x; show uniqueness by assuming two solutions and deriving they are equal.)

Q41. *[3 marks]* Determine whether the following is a valid argument. If valid, state the rules used; if invalid, name the fallacy:

Premises: "If the server is down, the website is offline." "The website is offline."
Conclusion: "The server is down."

Q42. *[4 marks]* Prove by contradiction: There are infinitely many prime numbers.
(Assume finitely many primes $p_1, \ldots, p_n$; form $N = p_1 p_2 \cdots p_n + 1$ and derive a contradiction.)

Logic, Predicates, Normal Forms & Circuits Q43–Q50

Q43. *[3 marks]* Construct a complete truth table for $F = (p \to q) \leftrightarrow (\neg p \vee q)$.
What do you observe about F? Name this property.

Q44. *[3 marks]* Simplify the following expression using logical equivalence laws. Show each step and the law used:

$$\neg(\neg p \vee q) \vee (p \wedge \neg r)$$

Q45. *[3 marks]* Let $D(x) =$"x is a dog", $F(x) =$"x has fur", $B(x) =$"x barks". Translate each into predicate logic and then negate (push $\neg$ all the way in):
(a) Every dog has fur.
(b) Some dog does not bark.
(c) No dog both has fur and barks.

Q46. *[3 marks]* Consider the domain of integers. Determine whether each statement is TRUE or FALSE, with brief justification:
(a) $\forall x\, \exists y\, (x \cdot y = 1)$
(b) $\exists x\, \forall y\, (x + y = y)$
(c) $\forall x\, \forall y\, \exists z\, (x + y = z)$

Q47. *[4 marks]* Convert $F = p \to (q \wedge r)$ to both CNF and DNF. Show all steps clearly.

Q48. *[4 marks]* Minimise the following function using a K-Map:

$$F(A, B, C, D) = \sum m(0, 1, 4, 5, 6, 7, 8, 9)$$

Draw the K-Map, identify all groups, and write the minimised Sum of Products (SOP) expression.

Q49. *[3 marks]* Design a logic circuit for the output $F = (A \oplus B) \wedge C + \bar{A}$.
(a) List all gates needed.
(b) Draw the circuit diagram (boxes labelled with gate type are acceptable).
(c) Evaluate F when $A = 1, B = 0, C = 1$.

Q50. *[4 marks]* Construct a formal proof of the following argument using rules of inference. State which rule is applied at each step:

Premises:

(1) $p \to q$

(2) $q \to r$

(3) $\neg r \vee s$

(4) p

Conclusion: s

Answer Key & Marking Hints

Full Solutions

Answer Key & Hints

MCQ ANSWERS (Part 2)

Q1: (c)	Q2: (b)	Q3: (b)	Q4: (b)	Q5: (c)	Q6: (c)	Q7: (b)	Q8: (b)	Q9: (b)	Q1
Q11: (c)	Q12: (b)	Q13: (c)	Q14: (b)	Q15: (a)	Q16: (c)	Q17: (a)	Q18: (a)	Q19: (b)	Q2
Q21: (b)	Q22: (b)	Q23: (b)	Q24: (c)	Q25: (b)					

Answer Key & Hints

Written Question Hints (Part 3)

Q26: (a) $A = \{-3, -2, -1, 0, 1, 2, 3\}$; (b) $A \cap B = \{-1, 0, 1, 2, 3\}$, $A \setminus B = \{-3, -2\}$; (c) No, $-3 \in A$ but $-3 \notin B$.

Q27: $A \setminus (B \cup C) = A \cap \overline{(B \cup C)} = A \cap (\bar{B} \cap \bar{C}) = (A \cap \bar{B}) \cap (A \cap \bar{C}) = (A \setminus B) \cap (A \setminus C)$

Q28: $\mathcal{P}(\{1, 2, 3\}) = \{\emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$; 8 elements $= 2^3$.

Q29: (a) $35 + 28 - 15 = 48$; (b) $(35 - 15) + (28 - 15) = 20 + 13 = 33$; (c) $60 - 48 = 12$.

Q30: Prove TRUE. Take $(a, (b, c)) \in LHS$: means $a \in A$ and $(b, c) \in B \cap C$, i.e. $b \in B$ and $c \in C$. So $(a, b, c) \in (A \times B)$ and $(a, b, c) \in (A \times C)$, hence in RHS. Reverse similarly.

Q31: $R = \{x \mid x \notin x\}$ — if $R \in R$ then $R \notin R$ and vice versa. Contradiction. Shows "any property defines a set" is inconsistent. ZFC uses restricted comprehension: $\{x \in A \mid P(x)\}$ — sets built from existing sets only.

Q32: $f(n) = \begin{cases} n/2 & n \text{ even} \\ -(n+1)/2 & n \text{ odd} \end{cases}$. Maps $0 \mapsto 0, 1 \mapsto -1, 2 \mapsto 1, 3 \mapsto -2, \ldots$ Injective: distinct n give distinct values. Surjective: every $z \in \mathbb{Z}$ is $f(2z)$ if $z \geq 0$ or $f(-2z - 1)$ if $z < 0$.

Q33: Assume list $s_1, s_2, \ldots$ of binary strings. Build d with $d_i = 1 - s_i[i]$ (flip diagonal bit). Then d differs from s_i at position i for each i, so d is not on the list. Contradiction.

Q34: (a) Countable — polynomials encoded as finite tuples of integers; (b) Countable — bijection with $\mathbb{N}^2$; (c) Uncountable — these are infinite binary strings, uncountable by diagonalization.

Q35: $n = 3k + 1 \Rightarrow n^2 = 9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1 \equiv 1 \pmod 3$. ✓

Q36: Contrapositive: n odd $\Rightarrow n^3$ odd. $n = 2k+1 \Rightarrow n^3 = (2k+1)^3 = 8k^3+12k^2+6k+1 =$

$2(\ldots)+1$, odd.

Q37: Assume $\sqrt{3} = p/q$, $\gcd(p,q) = 1$. Then $3q^2 = p^2 \Rightarrow 3 \mid p^2 \Rightarrow 3 \mid p$. Write $p = 3m$: $3q^2 = 9m^2 \Rightarrow q^2 = 3m^2 \Rightarrow 3 \mid q$. Contradicts $\gcd(p,q) = 1$.

Q38: $n = 2k$: $n^2 + n + 1 = 4k^2 + 2k + 1 = 2(2k^2 + k) + 1$ (odd). $n = 2k + 1$: $4k^2 + 4k + 1 + 2k + 1 + 1 = 4k^2 + 6k + 3 = 2(2k^2 + 3k + 1) + 1$ (odd). Both cases odd.

Q39: $a \mid b \Rightarrow b = as$; $a \mid c \Rightarrow c = at$. Then $mb + nc = mas + nat = a(ms + nt)$. Since $ms + nt \in \mathbb{Z}$, $a \mid (mb + nc)$.

Q40: Existence: $x = 5$ satisfies $15 - 6 = 9$. Uniqueness: $3x_1 - 6 = 9$ and $3x_2 - 6 = 9 \Rightarrow 3x_1 = 3x_2 \Rightarrow x_1 = x_2$.

Q41: INVALID. Fallacy of affirming the consequent. $p \to q$ and q does NOT give p. (Server down $\to$ offline; offline $\therefore$ server down — wrong, other reasons possible.)

Q42: Assume primes $p_1, \ldots, p_n$. Let $N = p_1 \cdots p_n + 1 > 1$. Let p be a prime factor of N. Then $p \mid N$ and $p \mid p_1 \cdots p_n$, so $p \mid (N - p_1 \cdots p_n) = 1$. So $p \leq 1$, but primes are ≥ 2. Contradiction.

Q43: All 4 rows of the biconditional are T (it's a tautology) since $p \to q$ IS defined as $\neg p \vee q$.

Q44: $\neg(\neg p \vee q) \vee (p \wedge \neg r) = (p \wedge \neg q) \vee (p \wedge \neg r) = p \wedge (\neg q \vee \neg r) = p \wedge \neg(q \wedge r)$

Q45: (a) $\forall x(D(x) \to F(x))$; neg: $\exists x(D(x) \wedge \neg F(x))$. (b) $\exists x(D(x) \wedge \neg B(x))$; neg: $\forall x(D(x) \to B(x))$. (c) $\forall x(D(x) \to \neg(F(x) \wedge B(x)))$; neg: $\exists x(D(x) \wedge F(x) \wedge B(x))$.

Q46: (a) FALSE — e.g. $x = 2$ has no integer y with $2y = 1$. (b) TRUE — $x = 0$: $0 + y = y$. (c) TRUE — just take $z = x + y \in \mathbb{Z}$.

Q47: $p \to (q \wedge r) \equiv \neg p \vee (q \wedge r)$. **DNF:** $\neg p \vee (q \wedge r)$ (already DNF). **CNF:** distribute: $(\neg p \vee q) \wedge (\neg p \vee r)$.

Q48: Plot minterms 0,1,4,5,6,7,8,9 on 4-var K-Map. Group $\{0,1,4,5,8,9\}$ (6 cells — not power of 2; use two groups): Group $\{0,1,8,9\}$: $\bar{B}\bar{D}$... best coverage: $\{0,1,4,5\} = \bar{A}\bar{B}$; $\{4,5,6,7\} = \bar{A}B$; $\{0,1,8,9\} = \bar{B}\bar{D}$... Final SOP: $F = \bar{A} + \bar{B}\bar{D}$.

Q49: (a) Gates: XOR, AND, NOT, OR. (b) XOR(A,B)$\to$AND with C; NOT(A); OR both results. (c) $A = 1, B = 0$: $A \oplus B = 1$; $(1) \wedge 1 = 1$; $\bar{A} = 0$; $F = 1 \vee 0 = 1$.

Q50: (1) $p \to q$ (premise); (2) $q \to r$ (premise); (3) By Hyp. Syllogism on (1),(2): $p \to r$; (4) p (premise); (5) By Modus Ponens on (3),(4): r; (6) $\neg r \vee s$ (premise); (7) By Disjunctive Syllogism on $\neg r \vee s$ and r (as $\neg(\neg r)$): s. ■

All 3 topics covered · 3 revision pages · 50 questions · Full answer key

You are ready. Good luck on your exam! ★★★